

Moller scattering 3

The Casimir trick uses matrix arithmetic to sum over spin states.

Let

$$\begin{aligned}f_{11} &= 8s(s - 8m^2) + 8u(u - 8m^2) + 192m^4 \\f_{12} &= -8s(s - 8m^2) - 96m^4 \\f_{22} &= 8s(s - 8m^2) + 8t(t - 8m^2) + 192m^4\end{aligned}$$

such that

$$\begin{aligned}\sum_{abcd} \mathcal{M}_t \mathcal{M}_t^* &= \frac{e^4}{t^2} f_{11} \\ \sum_{abcd} \mathcal{M}_t \mathcal{M}_u^* &= \frac{e^4}{tu} f_{12} \\ \sum_{abcd} \mathcal{M}_u \mathcal{M}_u^* &= \frac{e^4}{u^2} f_{22}\end{aligned}$$

By the Casimir trick we have

$$\begin{aligned}f_{11} &= \text{Tr} \left((\not{p}_4 + m) \gamma^\mu (\not{p}_2 + m) \gamma^\nu \right) \text{Tr} \left((\not{p}_3 + m) \gamma_\mu (\not{p}_1 + m) \gamma_\nu \right) \\f_{12} &= \text{Tr} \left((\not{p}_4 + m) \gamma^\mu (\not{p}_2 + m) \gamma^\nu (\not{p}_3 + m) \gamma_\mu (\not{p}_1 + m) \gamma_\nu \right) \\f_{22} &= \text{Tr} \left((\not{p}_3 + m) \gamma^\mu (\not{p}_2 + m) \gamma^\nu \right) \text{Tr} \left((\not{p}_4 + m) \gamma_\mu (\not{p}_1 + m) \gamma_\nu \right)\end{aligned}$$

Recall that slash matrices are formed as

$$\not{p} = p^\mu g_{\mu\nu} \gamma^\nu = p^0 \gamma^0 - p^1 \gamma^1 - p^2 \gamma^2 - p^3 \gamma^3$$

Eigenmath script